

Flexible Work Arrangements, Labor Supply and Firm Outcomes

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May 8, 2025

Abstract

This paper studies the impact of regular labor supply and increased worker coordination on worker and firm level productivity in the context of a large apparel manufacturing cluster in India. The status quo work arrangement in these firms is that workers (tailors) choose their own work timings, despite clear gains to both workers and firms of coordinating work times. Such flexibility slows the overall speed of production by limiting the extent of task specialization among workers. I work with a sample of 60 small firms and 420 workers, and incentivize all workers in treatment firms to come to work at the same time for a period of 4-weeks. Workers in treatment firms either receive a high or a low incentive, allowing me to estimate workers' willingness to pay (WTP) for flexibility by observing daily compliance at different bonus amounts. I collect detailed data on daily production at the worker and firm level, which allow me to estimate the effects of increased worker coordination on firm-level outcomes such as output, profits, revenues, costs, and order characteristics. I find that incentives increase compliance in treatment firms by 27-29% relative to control firms, but there is no effect on worker attendance or daily work hours. Workers in treatment firms increase the total number of breaks during the day, but there is no effect on the total duration of breaks. These changes in worker's labor supply add up to a significant improvement in worker coordination at the firm level: dispersion in worker arrival times decreases by 18-66% based on several measures of dispersion.

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1 Introduction and Motivation

A recent literature emphasizes the role of workers’ preferences for flexibility in explaining the prevalence of self-employment or casual-work over regular wage-work in developing countries (Cefala, Kaur, Schofield, and Shamdasani, Cefala et al.; Oh, Shamdasani, and Goraya, Oh et al.; Ho, Jalota, and Karandikar, Ho et al.). For example, Cefala, Kaur, Schofield, and Shamdasani (Cefala et al.) document flexible work schedules as the second most important reason (after earnings per day) for choosing a casual job over a regular job. However, while flexibility is an important job attribute for workers, flexible work arrangements may be costly for firms. For example, in settings where the production process requires a high degree of coordination from workers (assembly line manufacturing), providing flexibility may reduce productivity by increasing worker/machine idle time, or increase fixed operating costs (e.g., electricity). In fact, providing flexibility may be even more costly for large firms due to decreasing returns to managerial span of control (Lucas, Lucas) – this can have broader implications on firms’ ability to scale their workforce.

In this paper, I study the impact of regular labor supply and increased worker coordination on worker and firm level productivity. I study the context of a large apparel manufacturing cluster in India which features many small-medium firms employing 5-15 workers (tailors).¹ Firms are organized as batch processing lines, where each worker performs a distinct operation on the garment in batches of 80-100 pieces. The organization of these firms reflects the trade-off between firm and worker preferences motivated above.

First, tailors are hired on piece rate contracts and organize as independent contractors in firms. This enables them to choose their work timings, instead of being required to follow uniform work shifts like their counterparts working in large/export firms.² Effectively, this means that workers

¹These firms account for $\approx 30\%$ of all 2+ worker firms in this district and 3-digit industry group (Economic Census, 2013)

²In another ongoing project in Tirupur, I document large and systematic differences in work arrangements by firm size – see [Table A1](#). Interviews with both small and large firm owners suggests that differences in the provision of flexibility arise from a combination of institutional and technological constraints: (a) labor laws and social compliance from international buyers require large/export firms to systematically record work shifts in order to provide workers government benefits and overtime pay, and (b) large/export firms are more likely to organize production in assembly lines, while small firms follow batch production processes. Although both forms of production involve a high degree of task specialization among workers (i.e, workers perform only 1 distinct task per product/day), assembly lines have a stronger requirement for worker coordination than batch production processes.

do not coordinate their work schedules – in data from 60 firms and 420 workers, the difference in arrival times from company’s stated start time ranges from 60 minutes early to 75 minutes late (90-10 gap, see [Figure 2](#)) . Moreover, because workers are only incentivized on individual output as opposed to team output (e.g. only 100 sleeves + 100 collars versus 100 pieces of the completed garment), workers’ individual choice of work schedules may create a coordination externality on both other workers and the firm owner.

On the other hand, the production technology implies that worker coordination may improve production - because operations need to be performed sequentially to produce the final output, the lack of coordination in worker’s arrival times can either (a) limit worker’s ability to specialize on 1 task because they need to fill in for a late worker who usually performs the preceding task, or (b) increase idle time if they need to wait for the preceding task to be completed. Thus, such flexible work arrangements may slow the overall speed of production, and lower productivity by limiting specialization. This is consistent with owner’s beliefs about how flexibility affects production, and why imposing fixed work schedules may increase production (see [Figure 3a](#) and [Figure 3b](#)).

This project asks two questions: (1) how costly is it for firms to incentivize workers to coordinate work timings – in other words, how much do workers value flexibility in work schedules? and (2) Does increased worker coordination improve worker and firm outcomes? To answer these questions, I work with a sample of 60 small firms and their workers in the district, where workers in treatment firms are incentivized to follow fixed-work schedules for a 4-week period. Workers in control firms receive unconditional incentives for the same period, matched to a randomly chosen treatment firm each week to address concerns about income effects on worker productivity.

I find that treatment significantly increases the probability that workers arrive at work at or before owners’ preferred start times as elicited at baseline. This effect is large - the fraction of worker days that workers are present and arrive before the preferred time increases by 13 percentage points, relative to a control mean of 52% when incentives are active. However, this increase is only driven by workers who are present at the firm on that day, there is no effect on worker attendance. This effect is almost entirely driven by women, and workers who consistently arrived after the preferred start time at baseline. Workers in treatment firms do not reduce total daily work hours or increase

the total duration of breaks to compensate for arriving early. There is suggestive evidence that workers in treatment firms increase the number of breaks taken during the day. However, this increase is statistically insignificant.

This paper makes contributions to 3 strands of the literature: First, I experimentally estimate workers’ demand for flexible work arrangements. While this has been widely documented across many contexts (Mas and Pallais, b; Ho, Jalota, and Karandikar, Ho et al.; Baysan, Dar, Emerick, Li, and Sadoulet, Baysan et al.; Wiswall and Zafar, Wiswall and Zafar), this project’s contribution is to link worker’s WTP for flexibility with firm outcomes. This result will highlight an important, yet under-explored relationship between the provision of flexible work arrangements, and labor supply preferences (Mas and Pallais, a). Second, I build on previous work documenting search and hiring frictions in low-income countries (De Mel, McKenzie, and Woodruff, De Mel et al.; Hardy and McCasland, Hardy and McCasland; Bassi and Nansamba, Bassi and Nansamba; Caria and Falco, Caria and Falco; Garlick, Carranza, and Orkin, Garlick et al.) by documenting that even conditional on forming matches, firms may face barriers to coordinating workers. This could have wider economic consequences on firms’ ability to scale their workforce as coordination becomes costlier with scale (Lucas, Lucas). Finally, this project will contribute to our understanding of the internal organization of small, informal firms in developing countries, and the role of underlying market frictions, building on recent work such as Bassi, Muoio, Porzio, Sen, and Tugume (Bassi et al.); Bassi, Lee, Peter, Porzio, Sen, and Tugume (Bassi et al.). Specifically, this project will document worker preferences as the source of coordination costs for SMEs in developing countries, and how they affect firms’ ability to specialize.

The remainder of the paper is organized as follows: [section 2](#) describes the study context and sample firms, [section 3](#) describes the experimental design, implementation and data collection, and [section 4](#) lays out the empirical strategy. [section 5](#) presents the main results.

2 Study Context and Sample

The context of my study is Tiruppur district in the state of Tamil Nadu, India. This region is part of a major garment manufacturing hub in India, catering to both export and domestic markets

since the 1970s. The production process in Tiruppur is highly decentralized and carried out by sub-contracted units, called job-work units - these units will form my study sample. Job work units are *small* factories, employing about 10-30 workers, at least 50% of whom are women.³

Firms are organized as batch processing lines, where each worker performs a distinct operation on the garment in batches of 80-100 pieces. Operations need to be performed sequentially in order to produce a finish piece of garment. Firms typically specialize in broad product categories - such as t-shirts, bottomwear, innerwear, or kids wear. The average weekly order size in these firms is about 4,120 pieces, varying with product type/size.

The study takes place in 4 neighborhoods located in Tirupur North sub-district, Tirupur, Tamil Nadu. These neighborhoods were identified as high density clusters of small-medium job-work units during prior scoping work and area mapping exercises. I used the following criteria to screen firms to be part of the sample:

1. Has a stitching section
2. Employs at least 2 tailors who are not related to the firm owner or who work for pay (piece rate contract)
3. Must deliver finished products back to the buyer/retailer
4. Must receive cut-pieces of cloth to be stitched (ie does not only produce own designs)

All firms that satisfy these criteria were eligible for baseline. In case eligible firms were located adjacent to each other, I approached alternate firms in order to minimize spillovers between treatment and control firms located next to each other.

3 Experiment Design

Firms are randomly assigned to one of two treatment groups - (T) Treatment ($n = 30$), and (C) Control ($n = 30$). The treatment group was further randomized into (T1) high and (T2) low incentive sub-treatments, with daily bonus amounts set at Rs. 100 (high) and Rs. 50 (low)

³Note that these firms are of course larger than the typical small developing country firm; however they employ a substantial fraction of the workforce within the textile industry in Tiruppur district - as of 2013, the share of total employment in textiles in 10-30 worker firms ranged between 20-30% in rural and urban areas in Tiruppur district (source: Economic Census, 2013-14).

respectively. These amounts are approximately 15% and 7.5% of worker’s average daily earnings and were chosen based on owners’ estimates of the minimum amount needed to encourage workers to come to work early. Workers’ average daily earnings are Rs. 600. Treatment incentives are conditional on the worker arriving at the firm and starting work by the company’s preferred start time. Thus, the incentive encourages both on-time arrival and attendance. The total incentive amount was paid to workers as part of their weekly earnings. The unconditional incentive amount for control group firms was matched to the total realized payment made to a randomly selected firm from treatment group T2 (low incentive firm). These treatment-control firm (randomized) pairings were created separately by week.

3.1 Implementation and Data Collection

The experiment was implemented in two phases - Phase 1 was implemented with 17 firms (9 treatment, 8 control) from November 2024 to January 2025. Phase 2 was implemented with 43 firms (22 treatment, 21 control) from March 2025 to May 2025. Phase 1 only included high treatment firms (T1), and the half of the control firms did not receive any incentive (pure-control). In Phase 2, all control firms receive unconditional incentives matched to a randomly chosen treatment firm, up to the maximum allowable amount in low-treatment firms. These changes were made to simplify field logistics (minimizing distance between sample firms as well as firm attrition under limited budget for survey operations), and limited budget for survey incentives. **Figure 1** shows the study timeline.

Randomization for Phase 1 was conducted by generating a random order of treatment and control assignments, to which firm were matched in the order in which they were baselined. Randomization for Phase 2 was stratified on the degree of dispersion in worker arrival times around the owner’s preferred work start time, using worker’s usual arrival times as reported by the owner at baseline.

At baseline, I conduct a baseline survey with owners to capture owner and firm characteristics, a complete worker roster, and an order roster recording the task structure of each product that the firm is currently producing, along with piece rates received by both owners and workers (task-wise). Starting 1-2 weeks after baseline, I start collecting daily data for a period of 6-weeks,

out of which workers are paid incentives for 4-weeks.⁴ Each day, I record the worker’s attendance, arrival time, break times during the day, and daily output (by product-task). Every week, I also record worker productivity at their current task. Finally, in endline surveys I capture firm owner’s and worker’s time-use during and outside of work, and preferences for flexible work arrangements. These questions allow me to examine (1) the nature of workers’ commitments/restraints that drive their preference for flexibility, (2) how compliance with the incentive impacted these commitments/restraints, and (3) whether compliance shifted workers’ preferences for flexible (versus strict) work arrangements. Randomization and treatment assignment was done after baseline.

Baseline firm and worker characteristics, and balance by treatment arms are reported in [Table 1](#) and [Table 2](#) respectively. On average, firms hire 7 tailors and have been operating for 7.4 years. On average, firms produce 4,120 pieces of output each week, and earned a profit of INR 43,000 in the last 30 days (USD 490). Owners directly supervise workers (that is, without a manager/contrator) in almost all firms. About 70% owners are male, 52% have completed secondary school and have worked for about 18 years in the garment industry. While all firms perform “job-work” for external units, 28% control firms (3% treatment firms) also produce their own orders. Correspondingly control firms have There are some notable differences between control and treatment firms at baseline – for instance, treatment firms are 2 years younger, less likely to produce export orders, and earn INR 2,100 less in revenues per worker (INR 1760 less in profits per worker). However, all differences except the probability of producing owner orders are statistically insignificant.

Worker characteristics are balanced across treatment and control firms ([Table 2](#)). Workers are on average 36 years old, 40% are male and most workers are married (93%) and have a child younger than 14 years of age (86%). 84% workers live within a 15 minute commute to the firm, have worked at the firm for 2-3 years and have 14 years of experience in the garment industry.

4 Empirical Strategy

Given the panel set up of this RCT, I plan to use both analysis of co-variance (ANCOVA) and difference-in-differences (DD) specifications to estimate treatment effects (following best practices

⁴In Phase 1, I collected data 1 week pre-incentives and 1 week post-incentives. In Phase 2, I am collecting 2 weeks of pre-incentives data.

recommended Bruhn and McKenzie (Bruhn and McKenzie); Burlig, Preonas, and Woerman (Burlig et al.). The specification for individual level outcomes is as follows:

$$Y_{i,f,t} = \alpha + \beta T_{i,f} + \mu_w + \theta Y_{i,f;t=0} + \mathbf{X}'_{i,f} \gamma + \varepsilon_{i,f,t} \quad (1)$$

$$Y_{i,f,t} = \alpha + \beta T_{i,f} \times \mathbb{I}\{\text{During}\}_t + \mu_w + \lambda_{i,f} + \varepsilon_{i,f,t} \quad (2)$$

where $Y_{i,f,t}$ is the outcome of interest for worker i in firm f at time t , where time is either day or week, depending on the outcome variable. $T_{i,f}$ is an indicator that equals 1 if a firm f is assigned to treatment (fixed over time). μ_w are study-week fixed effects.⁵ Standard errors $\varepsilon_{i,f,t}$ are clustered at the firm level.

In Equation 1, $Y_{i,f;t=0}$ denotes the baseline value of the outcome variable (worker level) and $\mathbf{X}'_{i,f}$ denotes a vector of controls for study phase and randomization strata.⁶ Equation 1 only includes incentives weeks 1-4. Equation 2 uses all 6-weeks of data, $\mathbb{I}\{\text{During}\}_t$ is an indicator variable that takes the value 1 if the study week is an incentive week, and $\lambda_{i,f}$ denotes worker fixed effects, which absorb baseline controls.

For analyzing firm level outcomes, I will follow similar specifications, depending on the frequency at which an outcome variable was measured:

⁵Study weeks take the value 1 to 4 for the 4 incentive weeks, 0 for the week prior to incentives. Weeks 0-4 are common across Phase 1 and 2. The first pre-incentive week in Phase 2 is coded week -1, and the final, post-incentive week in Phase 1 is coded week 5. In the current analysis, I drop the post-incentives week (week 5).

⁶Randomization in Phase 2 was stratified on owner-self-reported baseline level of dispersion in worker arrival times around the owner's preferred start-time. Ex-ante, I expect this measure to affect worker compliance, and thus further downstream effects, and thus increase the power of my tests. Ex-post, if this measure is not related to individual or firm outcomes, including strata fixed effects will reduce the power of my tests; in this case, I will not include strata fixed effects in my regressions and will only include study phase fixed effects.

$$Y_{f,t} = \alpha + \beta T_f + \mu_w + \theta Y_{f;t=0} + \mathbf{X}'_f \gamma + \varepsilon_{f,t} \quad (3)$$

$$Y_{f,t} = \alpha + \beta T_f \times \mathbb{I}\{\text{During}\}_t + \mu_w + \lambda_f + \varepsilon_{f,t} \quad (4)$$

where indices and variables are defined as above, and standard errors $\varepsilon_{f,t}$ are clustered at the firm level. Some firm outcomes are only measured at endline, in which case I will run [Equation 3](#) without the t subscript and week fixed effects.

5 Results

5.1 Worker Outcomes: Labor Supply

First, I test the effects of the incentives on worker compliance. Columns (1)-(4) in [Table 3](#) report results for worker compliance and attendance using both the ANCOVA and difference in differences specification (odd and even columns respectively). Compliance is defined as an indicator for being present at the firm, and arriving by the owner’s preferred start time. Incentives significantly increase compliance among workers by 12-13 percentage points, a 25-27% increase relative to workers in control firms (columns (1) and (2)). This is not driven by any changes in worker attendance (columns (3) and (4)) - the treatment effect on attendance is 0 percentage points. The effects on compliance are larger, and more meaningful when conditioned on workers being present at the firm – on average, attendance at control firms during (before) the incentive weeks is 69% (61%). Among workers who are present at the firm, incentives increase compliance significantly by 17 percentage points. Relative to the control mean of 69%, the treatment increases the probability of workers arriving by the owner’s preferred time to 86%.

Next, I examine whether these effects vary by worker characteristics. I focus on two dimensions: gender, and whether the worker consistently arrived after the preferred start time during the pre-incentives weeks (hereafter, referred to as “late” versus “early” workers). [Table 4](#) reports heterogeneity results using the difference in differences specification. Compliance is driven by female

workers, and workers who were consistently late at baseline – female workers are 13 percentage points more likely than male workers to comply (column 1); “late” workers are 27 percentage points more likely than “early” workers to comply (column 3). Treatment effects on attendance for women and “late” workers smaller than the respective base categories, however the differences are small in magnitude and insignificant. This is consistent with the result above that incentives did not change worker attendance in treatment firms.

Do workers compensate for arriving earlier at the firm by working fewer hours, or taking more breaks? Next, I look at treatment effects on daily work hours, the number and duration of breaks taken. Daily work hours is defined as the total hours worked in a day, excluding the duration of breaks. Breaks include any time taken for lunch, tea, or other reasons for the worker had to leave the firm. All outcomes are coded 0 on days that a worker was absent. Table 5 reports these results for both the ANCOVA and DD specifications. Workers in treatment firms work 0.13-0.15 hours, or 7.8-9 minutes, more than workers in control firms (columns 1-2 of Table 5). This is an insignificant, 2% increase relative to the control mean of 6.30-6.89 daily hours. However, I find a large, but insignificant increase in the total number of breaks taken by workers in treatment firms using the DD specification (column 4) – pre-incentives, workers in control firms took an average of 2.4 breaks during the day. During incentive weeks, workers in treatment firms take an additional 1.5 breaks during the day – a 62% increase relative to control firms. On the other hand, treatment effects on total duration of breaks (in minutes) is small and insignificant.

These results mask interesting heterogeneity by worker characteristics. Table 6 reports heterogeneity in difference-in-differences effects by gender (columns (1)-(3)) and “late” versus “early” workers (columns (4)-(6)). The small increase in worker hours is driven by female, and “late” workers – both types of workers increase work hours by approximately 30 minutes, while treatment effect for male or “early” workers is close to 0 (columns (1) and (4)). Relative to control means during the pre-incentive weeks, female workers in treatment firms catch up with male workers in terms of work hours in treatment firms. On the other hand, treatment widens the gap in work hours between “late” and “early” workers. Similarly, treatment effects on the total number and duration of breaks are driven by female workers; however, these estimates are extremely noisy (columns (2) and (3)).

5.2 Firm Outcomes: Coordination in Worker Arrival Times

Finally, I examine how treatment affects worker coordination at firms, conditional on being present. **Table 7** reports treatment effects for three measures of dispersion in worker arrival times at the firm-day level: minutes between the 1st and last worker to arrive at the firm (columns (1)-(2)), the 90-10 gap in worker arrival times (relative to the preferred start time) (columns (3)-(4)), and the deviation from preferred start time for the 90th percentile worker (columns (5)-(6)). This regression excludes firm-days on which only 1 worker was present at the firm, since the notion of coordination is not meaningful in such cases.

Total dispersion in worker arrival times is 120 minutes on average at baseline – treatment reduces dispersion by 24 minutes, a 20-22% decrease in dispersion, or improvement in coordination relative to control firms (columns (1)-(2)). The 90-10 gap is a more conservative measure of dispersion which excludes extreme outliers – workers who arrive extremely early or extremely late. For example, the 90-10 gap is 40 minutes lower than total dispersion in control firms during the pre-incentives week(s). Treatment further reduces the 90-10 gap by 13 minutes, a significant 18% decrease relative to the control mean of 69 minutes during the incentive weeks. However, this effect is not significant after adding firm fixed effects (column 4). Finally, I look at treatment effects on deviation for the 90th percentile worker (sorted by arrival time at the firm). This measure allows me to test whether these effects are driven by workers for whom the incentive was binding. Indeed, the deviation of the 90th percentile work reduces by 16-17 minutes in treatment firms, relative to the control group mean of 24 minutes during incentive weeks and 46 minutes pre-incentives. These effects amount to 36-66% reduction in dispersion, or improvement in coordination as a result of treatment.

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6 Figures

Figure 1: Study Timeline

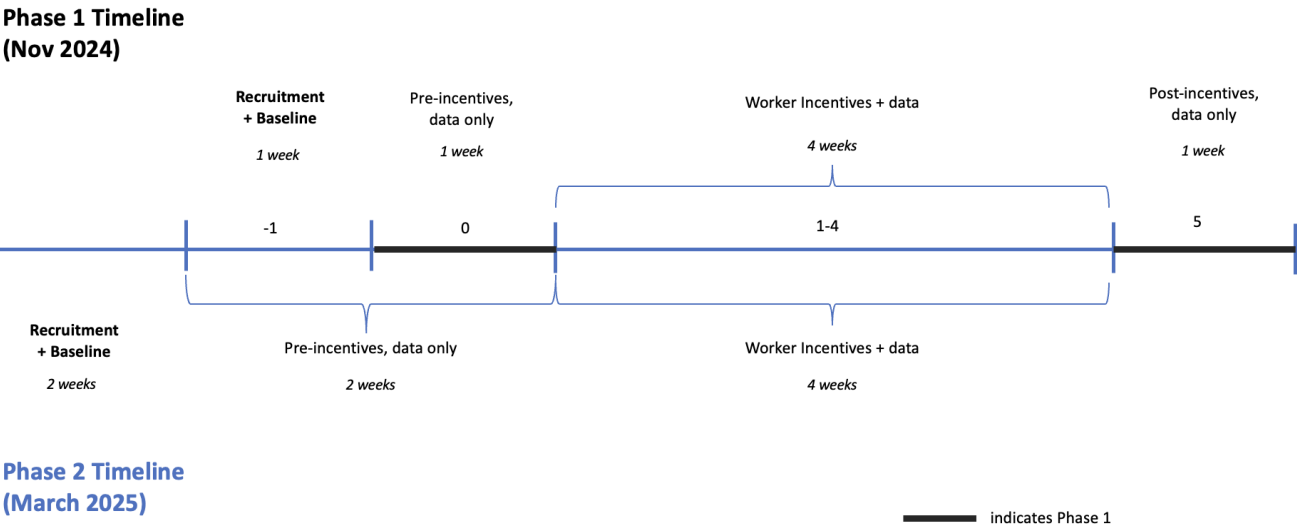
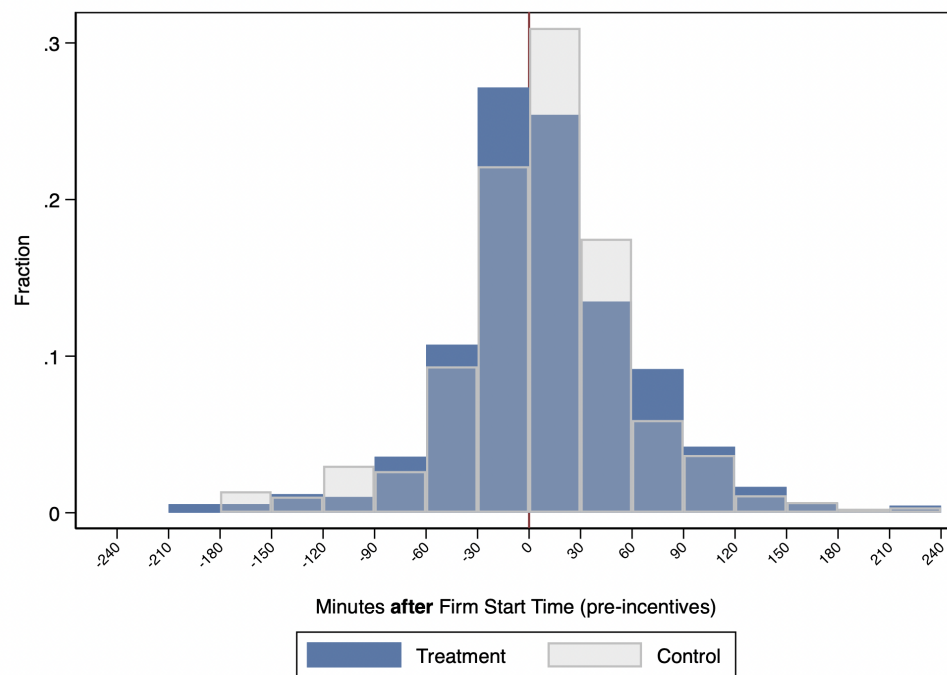
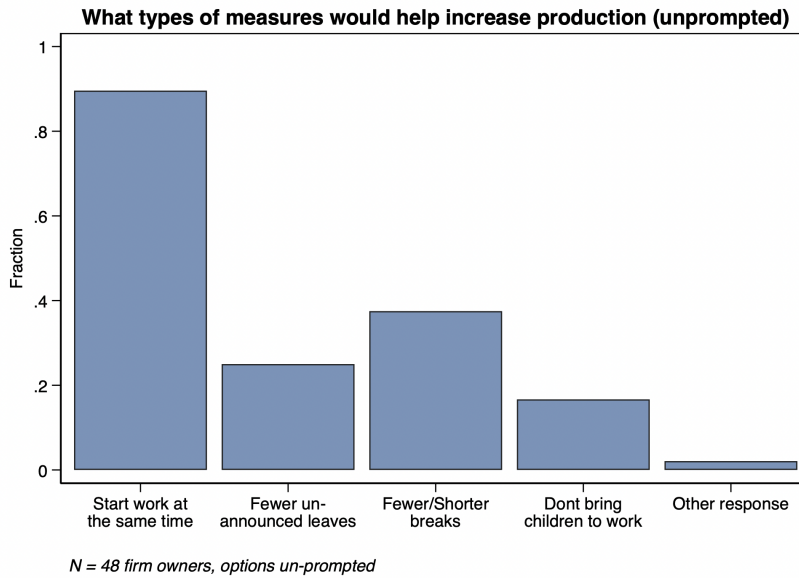


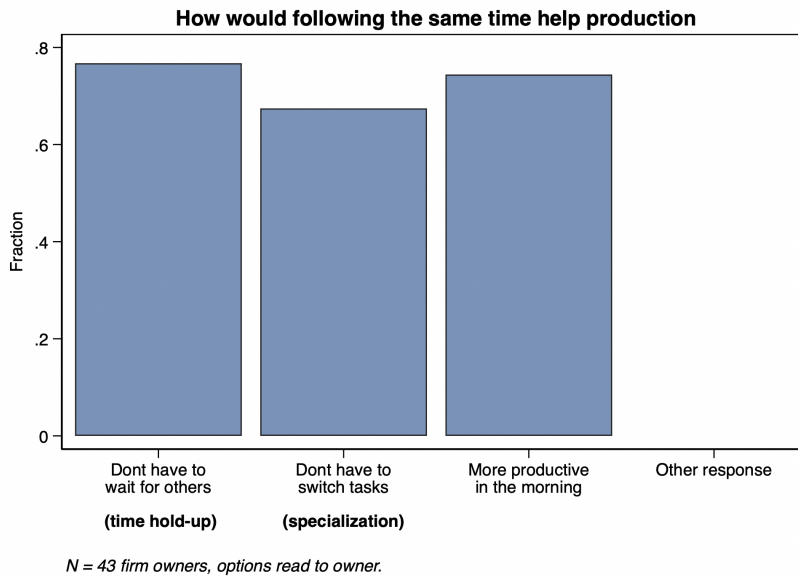
Figure 2: Dispersion in Worker Arrival Times, before Incentives



Notes: This figure plots the distribution of the deviation of worker arrival times from the owner's preferred start time. N = 60 firms, 420 workers.



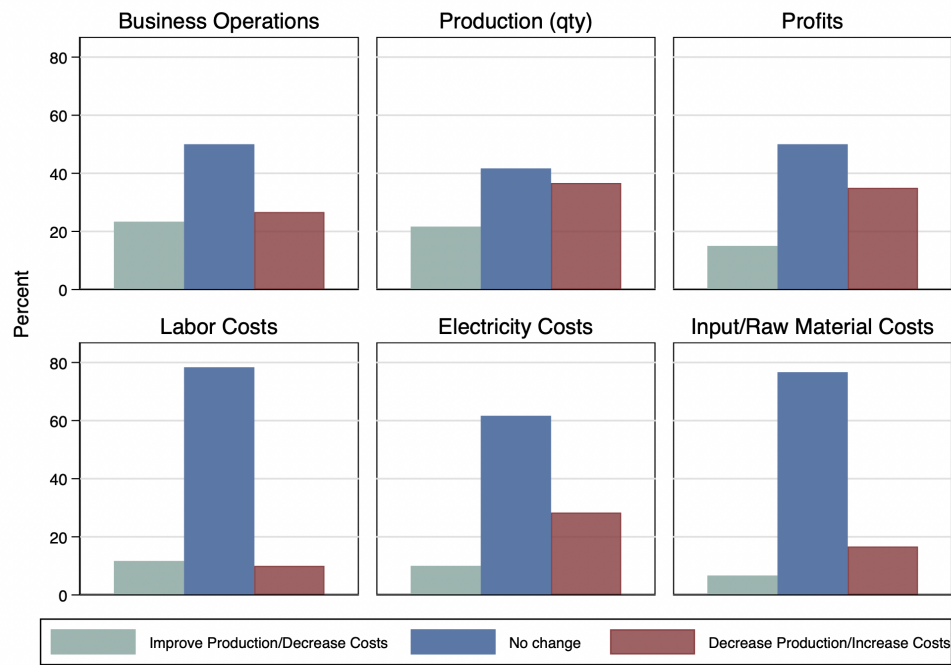
(a) Measures that could help increase production



(b) Why is coordination in worker arrival times important

Notes: Panel A reports owner's responses from the question "What type of measures would help increase production?", with options unprompted. $N = 60$. Panel B reports owner's responses from the question "If everybody starts work at the same time, how will it improve production?", with options 1 = Workers don't have to wait for others, 2 = Workers don't have to switch tasks, 3 = Workers are more productive during morning, and Others. This data is from the full sample of 60 firm owners.

Figure 4: How Providing Flexibility Affects Firm Operations



This figure reports owner's responses from the question: "Now I will list 4 different business objectives. For each of these, do you think providing workers flexibility affects this objective positively, negatively, or has no effect?" This data is from the full sample of 60 firm owners.

7 Tables

Table 1: Balance on Baseline Firm Characteristics

	Control		Treatment		T - C	p-value
	Mean	SD	Mean	SD		
Firm Characteristics						
# Tailors	7.14	(2.55)	7.03	(2.44)	-0.14	0.81
# Helpers	1.90	(1.45)	1.23	(1.06)	-0.69	0.03
Total # Machines	11.38	(4.52)	10.06	(3.03)	-1.35	0.17
Owner directly supervises workers	0.93	(0.26)	1.00	(0.00)	0.07	0.16
Firm Age (yrs)	7.38	(5.85)	5.30	(4.85)	-2.08	0.15
Produce own orders	0.28	(0.45)	0.03	(0.18)	-0.24	0.01
Export orders	0.14	(0.35)	0.06	(0.25)	-0.08	0.35
Only stitching section	0.66	(0.48)	0.71	(0.46)	0.06	0.60
Cutting section	0.80	(0.42)	0.22	(0.44)	-0.53	0.04
Finishing (ironing, packing)	0.50	(0.53)	0.11	(0.33)	-0.26	0.27
Weekly Output (Normal)	4120.00	(2346.45)	4128.57	(3232.34)	96.74	0.89
Weekly Output (Peak)	6256.00	(3508.45)	6882.14	(4460.53)	721.01	0.47
Weekly Output (Lean)	2616.00	(2031.97)	2417.86	(2289.11)	-131.57	0.81
Revenues (last 30 days)	147855.56	(137491.26)	120733.33	(107104.18)	-28212.87	0.32
Revenue per worker	19495.15	(14922.86)	17444.69	(13410.95)	-2151.14	0.53
Profits (last 30 days)	43163.26	(75533.87)	22966.67	(17097.05)	-20494.54	0.16
Profits per worker	5186.90	(5697.60)	3447.63	(2556.31)	-1761.13	0.13
Total Costs - wage bill (last 30 days)	31025.17	(21543.84)	32591.61	(21720.48)	1496.43	0.79
Owner Demographics						
Age	37.10	(7.87)	35.74	(6.97)	-1.31	0.50
Male	0.76	(0.44)	0.77	(0.43)	0.02	0.89
Completed Secondary School	0.52	(0.51)	0.39	(0.50)	-0.14	0.29
Commute Time (mins)	5.69	(6.33)	8.19	(9.62)	2.46	0.25
Experience in Industry (yrs)	18.83	(10.73)	16.70	(8.30)	-2.08	0.40
Observations	29		31			

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with robust standard errors.

Table 2: Balance on Baseline Worker Characteristics

	Control		Treatment		T - C	<i>p-value</i>
	Mean	SD	Mean	SD		
Age	36.62	(9.15)	36.48	(9.74)	-0.27	0.83
Male	0.44	(0.50)	0.41	(0.49)	-0.03	0.62
Married	0.93	(0.45)	0.91	(0.49)	-0.02	0.69
Has children < 14 yrs	0.86	(1.36)	0.84	(1.30)	-0.04	0.87
Has elderly member in HH	0.37	(0.48)	0.37	(0.49)	0.01	0.87
Related to owner	0.25	(0.43)	0.28	(0.45)	0.03	0.65
Native to TN	0.91	(0.29)	0.96	(0.20)	0.05	0.13
Commute < 15 mins	0.84	(0.37)	0.84	(0.36)	0.01	0.85
Completed Secondary School	0.46	(0.50)	0.40	(0.49)	-0.06	0.27
Experience in Industry (yrs)	14.70	(9.05)	14.34	(9.96)	-0.51	0.67
Tenure at Firm (yrs)	2.95	(4.03)	2.08	(3.07)	-0.88	0.16
Observations	189		187			

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with standard errors clustered at the firm level.

Table 3: Treatment Effects on Compliance and Attendance

	Compliance		Attendance		Compliance cond. on Attendance	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.13*** [0.04]		0.00 [0.04]		0.17*** [0.04]	
Treatment $\times$ During Incentives		0.12*** [0.04]		0.00 [0.04]		0.17*** [0.04]
Control Mean (Pre-Incentives)	.43	.43	.71	.71	.61	.61
Control Mean (Incentive Weeks)	.52	.52	.76	.76	.69	.69
Treatment Effect %	.24	.28	.01	0	.25	.28
Observations	7167	10948	7246	11025	5446	8154

Notes: All specifications include day-of-week, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. Odd columns report results from the ANCOVA specification in Equation 1, and controls for the average of the outcome variable during pre-incentive weeks. Even columns report results from the DD specification in Equation 2 with worker fixed effects. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. Attendance is defined as an indicator for whether a worker was present at the firm on that day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Heterogeneity in Treatment Effects on Compliance and Attendance

	x Female		x Late at BL	
	Compliance (1)	Attendance (2)	Compliance (3)	Attendance (4)
Treatment $\times$ During $\times$ Variable	0.13** [0.05]	-0.02 [0.04]	0.27*** [0.07]	-0.03 [0.06]
Treatment $\times$ During Incentives	0.05 [0.05]	0.01 [0.04]	-0.02 [0.05]	0.02 [0.04]
Control Mean (Pre-Incentives)	.43	.71	.43	.71
Interaction Variable Mean Pre-Incentives	.34	.34	.32	.32
Interaction Base Variable Mean Pre-Incentives	.6	.6	.55	.55
<i>p-val</i> T $\times$ During $\times$ Var = T $\times$ During	0.00	0.00	0.00	0.00
Observations	10331	10406	10948	11025

Notes: All specifications include day-of-week, week, phase $\times$ strata and worker fixed effects, from the specification in Equation 2. Standard errors are clustered at the firm level. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. Attendance is defined as an indicator for whether a worker was present at the firm on that day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Treatment Effects on Daily Work Hours

	Work Hours		Total # Breaks		Total Break Duration (mins)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.15 [0.41]		0.03 [0.20]		-1.22 [3.29]	
Treatment $\times$ During Incentives		0.14 [0.41]		1.50 [1.35]		-1.90 [3.21]
Control Mean Pre-Incentives	6.3	6.3	2.42	2.42	45.67	45.67
Control Mean Incentive Weeks	6.89	6.89	2.27	2.27	51.91	51.91
Treatment Effect %	.02	.02	.01	.62	-.02	-.04
Observations	6648	10022	3858	6030	7246	11030

Notes: All specifications include dow, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. Odd columns report results from the ANCOVA specification in Equation 1, and controls for the average of the outcome variable during pre-incentive weeks. Even columns report results from the DD specification in Equation 2 with worker fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Heterogeneity in Treatment Effects on Daily Work Hours

	x Female			x Late at BL		
	Work Hours	Total # Breaks	Total Break Duration (mins)	Work Hours	Total # Breaks	Total Break Duration (mins)
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During $\times$ Variable	0.29 [0.45]	2.38 [2.38]	-1.72 [3.41]	0.30 [0.61]	-2.61 [2.75]	0.06 [4.39]
Treatment $\times$ During Incentives	0.00 [0.45]	0.13 [0.17]	-0.98 [3.21]	-0.02 [0.48]	2.86 [2.76]	-1.93 [3.84]
Control Mean Pre-Incentives	6.3	2.42	45.67	6.3	2.42	45.67
Interaction Variable Mean Pre-Incentives	6.45	2.37	46.4	6.87	2.3	50.91
Interaction Base Variable Mean Pre-Incentives	7.11	2.48	51.1	5.74	2.56	40.06
<i>p-val</i> T $\times$ During $\times$ Var = T $\times$ During	0.00	0.00	0.00	0.00	0.00	0.00
Observations	9570	5906	10411	10022	6030	11030

Notes: All specifications include day-of-week, week, phase $\times$ strata and worker fixed effects, from the specification in Equation 2. Standard errors at clustered at the firm level. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. Attendance is defined as an indicator for whether a worker was present at the firm on that day. The sample for these regressions excldes days on which a firm was shut, and no production took place. Columns (1)-(3) test for heterogeneity in treatment effects by gender, with Males as the base category. Columns (4)-(6) test for heterogeneity in treatment effects by whether a worker was consistently late at baseline, with workers as the base category.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Treatment Effects on Coordination - Firm Level

	Mins between 1st and last worker		Mins between the 10th and 90th pctl worker		Deviation from start-time: 90th pctl worker	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	-23.91** [11.30]		-12.68* [6.74]		-15.93*** [4.87]	
Treatment $\times$ During Incentives		-24.75* [12.70]		-12.73 [8.18]		-16.47** [6.59]
Control Mean Pre-Incentives	124.22	124.22	82.95	82.95	45.86	45.86
Control Mean Incentive Weeks	110.71	110.71	68.83	68.83	24.14	24.14
Treatment Effect %	-.22	-.2	-.18	-.15	-.66	-.36
Observations	1138	1625	1138	1625	1138	1625

Notes: This table reports firm-day level outcomes on coordination in worker arrival times. All specifications include dow, week and phase $\times$ strata fixed effects. Odd columns report results from the ANCOVA specification in Equation 3, and control for the average of the outcome variable during pre-incentive weeks. Even columns report results from the DD specification in Equation 4 with firm fixed effects. Standard errors are clustered at the firm level. All outcome variables are in minutes, and are only computed for firm-days where strictly greater than 1 worker was present.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A Appendix Tables

Table A1: Flexibility by Firm Size

	Small Firms		Large Firms	
	Mean	SD	Mean	SD
Timings Same	0.49	(0.50)	0.77	(0.42)
Timings Strict	0.15	(0.36)	0.50	(0.51)
Entry time is recorded	0.03	(0.17)	0.50	(0.51)
Pay cut if late	0.06	(0.23)	0.40	(0.49)
Margin of Lateness				
Acceptable to be 5 minutes late	0.97	(0.17)	0.94	(0.24)
Acceptable to be ≤ 30 minutes late	0.91	(0.29)	0.69	(0.47)
Acceptable to be > 30 minutes late	0.85	(0.36)	0.62	(0.49)
Observations	70		48	

Notes: SD in parenthesis (even columns). Small firms are defined as firms with up to 20 workers, and where the owner is often both a tailor (either full or part-time) and manager of the firm. Large firms include firms with at least 50 workers, where ownership and factory-floor management are separate.